An Adaptive IDS Based on CIL for Continuously Evolving Threats Cybersecurity

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Abstract

This report presents an adaptive Intrusion Detection System (IDS) using Class-Incremental Learning (CIL) to tackle evolving cyber threats. We compare the performance of three replay-based strategies—Experience Replay (ER), Incremental Classifier and Representation Learning (iCaRL), and Dark Experience Replay++ (DER++), used to mitigate catastrophic forgetting in IDS. Experiments on the UNSW-NB15 and CICIDS-2017 datasets demonstrate that DER++ outperforms ER and iCaRL, providing better performance stability and knowledge retention in complex attack scenarios.

1 Introduction

Intrusion Detection Systems (IDS) are key tools for detecting cyber threats in network environments (Zhang et al., 2024). Most traditional IDS follow a static training paradigm, where models are trained offline on a fixed dataset and then deployed without further adaptation (Guo et al., 2025). This assumption does not hold in real-world settings, where network traffic and attack patterns change over time.

As threats evolve, static models fail to detect unseen attacks and require repeated full retraining. To address these limitations, IDS needs to incrementally acquire, update, accumulate, and exploit knowledge (Wang et al., 2024). Continual Learning (CL) provides a strategy for incrementally incorporate new knowledge while preserving previously acquired information. Class-Incremental Learning (CIL) is especially suitable for IDS, as it allows the model to sequentially learn new attack classes over time while dynamically expanding its classification space (Zhou et al., 2024). However, a central challenge in CIL is catastrophic forgetting, which occurs when updating the model with new data leads to performance degradation on previously learned attack patterns (Zhang et al., 2024; Guo et al., 2025; Zhou et al., 2024; Wang et al., 2024).

Catastrophic forgetting will be faced up using three representative replay-based methods: Experience Replay (ER), Dark Experience Replay++ (DER++), and Incremental Classifier and Representation Learning (iCaRL). These approaches provide complementary mechanisms for balancing stability and plasticity through memory rehearsal and representation preservation, making them well-suited for adaptive IDS applications. Experiments under multiple class-incremental scenarios will be conducted on cybersecurity datasets CICIDS-2017 and UNSW-NB15 and results will be compared. Our results show that DER++ provides more stable performance than ER and iCaRL under highly incremental intrusion detection scenarios with a fixed memory buffer.

2 Dataset

To evaluate our adaptive Intrusion Detection System (IDS) under a Class Incremental Learning (CIL) setting, we employ two publicly available and widely used cybersecurity benchmark datasets: **UNSW-NB15** (2015) and **CICIDS-2017** (2017). These datasets were selected due to their multi-class attack structure, realistic traffic generation methodology, and suitability for studying catastrophic forgetting in evolving cybersecurity environments.

2.1 UNSW-NB15 Dataset (2015)

The UNSW-NB15 dataset was developed in 2015 by the Australian Centre for Cyber Security (ACCS) at the University of New South Wales. It was generated using the IXIA PerfectStorm tool to simulate realistic network traffic containing both benign and malicious activities.

The dataset is provided with predefined training and testing splits:

- Training set: 175,341 samples

- Testing set: 82,332 samples
- Total features: 45 columns

Each sample describes network flow characteristics such as packet counts, byte statistics, time-based attributes, and protocol information.

The dataset contains nine attack categories in addition to normal traffic: *Fuzzers, Analysis, Backdoor, DoS, Exploits, Generic, Reconnaissance, Shellcode, and Worms*.

Two label related columns are provided:

- `label`: binary classification (0 = normal, 1 = attack)
- `attack_cat`: multi-class attack category

Since this work focuses on Class Incremental Learning, the `attack_cat` column is used as the target variable, as it enables sequential introduction of new attack classes.

Data Cleaning and Preprocessing (UNSW-NB15)

The following preprocessing steps were applied:

- The `id`, `label`, and `is_sm_ips_ports` columns were removed due to lack of semantic meaning or low information content;
- Duplicate rows were removed (none were found);
- The *Worms* class was removed due to its extremely small number of samples (130 instances), which is insufficient for stable incremental training;
- Categorical features (`proto`, `service`, `state`) were encoded using one-hot encoding.

After preprocessing, the dataset maintains a multi-class structure with inherent class imbalance, making it suitable for evaluating catastrophic forgetting in incremental settings.

2.2 CICIDS-2017 Dataset (2017)

The CICIDS-2017 dataset was developed by the Canadian Institute for Cybersecurity (CIC) and contains realistic enterprise network traffic captured over five days. It includes modern attack scenarios such as DDoS, PortScan, Botnet activity, brute-force attacks, web attacks, and infiltration attempts.

The dataset is distributed across eight CSV files corresponding to different capture days. These files were merged into a unified dataset before preprocessing.

Initially, the dataset contained:

- 2,830,743 samples
- 79 features

Data Cleaning and Preprocessing (CICIDS-2017)

The following steps were applied:

- Column names were standardized by removing extra whitespace;
- Infinite values were replaced with NaN;
- Rows containing missing values were removed;
- Attack labels were unified into consistent categories;
- Extremely rare classes were removed:
  - *Infiltration* (36 samples);
  - *Heartbleed* (11 samples);

  These classes were excluded because they do not provide enough samples for stable model learning in a CIL setting;
- Columns containing only zero values were removed, as they provide no predictive information;
- Duplicate rows were removed to avoid bias;
- Perfectly correlated features were identified and redundant features were eliminated.

After cleaning, the feature space was reduced from 79 to 64 features. The final dataset exhibits strong class imbalance, with benign traffic dominating the distribution. This imbalance reflects real network environments and makes the dataset particularly suitable for evaluating replay based continual learning strategies.

2.3 Relevance to Class-Incremental Learning

Both datasets are well suited for evaluating Class-Incremental Learning (CIL)-based intrusion detection due to several key characteristics. Firstly, they contain multiple attack categories, which is essential for testing how a model can learn and distinguish between various types of cyber threats. Additionally, both datasets exhibit significant class imbalance, which is a common challenge in real-world cybersecurity scenarios, where malicious traffic is often much less frequent than benign traffic.

These datasets include rare attack types that simulate emerging threats, allowing the model to be tested on its ability to recognize novel and infrequent attack patterns. They also support the simulation of sequential class introduction, which is critical for assessing how well the model can adapt to new attacks over time without forgetting previously learned classes.

Finally, the combination of a controlled benchmark dataset (UNSW-NB15) and a large-scale dataset (CICIDS-2017) enables the evaluation of iCaRL, Experience Replay, and DER++ strategies for mitigating catastrophic forgetting under both structured and complex traffic scenarios.

3 Methodology

We designed an adaptive Intrusion Detection System (IDS) with the Class-Incremental Learning (CIL) paradigm, a setting of continual learning in which new classes are introduced sequentially and the model must learn them without accessing the full past data distribution. This setting realistically reflects the evolving nature of cyber threats, where novel attack types emerge over time and retraining from scratch is computationally expensive and often impractical.

3.1 Problem Formulation

Given a sequence of tasks, where each task introduces a disjoint set of attack classes, the model is trained sequentially on each task without revisiting the entire historical dataset. The first task always includes the benign traffic class to provide an initial decision boundary between normal and malicious traffic. Subsequent tasks introduce new attack categories according to predefined scenarios.

The key challenge in this setting is *catastrophic forgetting*, i.e., the degradation in performance on previously learned classes after training on a new set of classes (a task). To mitigate this issue, we implemented and compared three replay-based strategies: iCaRL (Rebuffi et al., 2017), Experience Replay (ER) (Riemer et al., 2019), and Dark Experience Replay++ (DER++) (Buzzega et al., 2020).

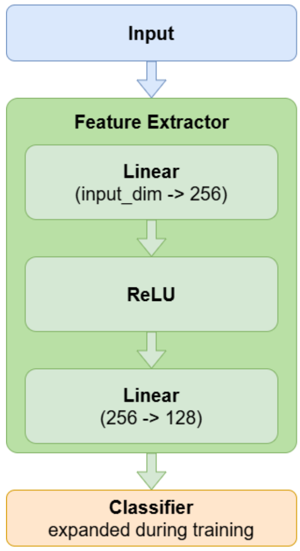


Figure 1: Model structure

3.2 Model Architecture

Given the tabular nature of network flow features, we adopted a lightweight fully connected neural network to ensure fast training and low computational overhead, both crucial for near-real-time intrusion detection. The model consists of:

- An input linear layer projecting feature vectors into a hidden representation layer;
- ReLU activation function;
- A second linear layer producing class logits.

To support class-incremental learning, the classifier layer is dynamically expandable: when new classes are introduced it is extended by adding new output neurons. The weights corresponding to previously learned classes are preserved and reused for the following task, enabling knowledge retention while learning new categories.

In addition, before training on each task, all feature vectors are normalized using up-to-normalization (also known as max normalization), where each feature is divided by its maximum observed value in the current task. This normalization ensures that all features are on a comparable scale, which stabilizes the training process.

3.3 Replay-Based Continual Learning Strategies

iCaRL (Incremental Classifier and Representation Learning) (Rebuffi et al., 2017) combines exemplar replay with knowledge distillation. For each class, a fixed number of representative samples (exemplars) are stored in a memory buffer using a herding strategy[1]. During training on new tasks, the model is optimized using a classification loss on current data and a distillation loss that preserves the output distribution of previous tasks.

ER (Experience Replay) (Riemer et al., 2019) maintains a memory buffer containing samples from previous tasks. At each training step, mini-batches are composed of both current-task samples and randomly drawn past samples. The model is trained with standard cross-entropy loss over this joint batch.

DER++ (Dark Experience Replay) (Buzzega et al., 2020) extends ER by storing not only input-label pairs but also the corresponding logits produced by the model at the time of storage. During replay, the model minimizes a mean-squared error between current and stored logits, preserving so-called "dark knowledge". DER++ is an advanced version of DER particularly well-suited to highly imbalanced datasets: by preserving soft targets through stored logits, it maintains inter-class relationships, including information about minority classes, helping prevent majority classes from dominating the learned representation.

3.4 Implementation Details

Both datasets were partitioned into multiple task sequences called scenarios: $Scenario_j = \{\mathcal{T}_1, ..., \mathcal{T}_k\}$, where each task $\mathcal{T}_i$ is a subset of attack classes, reflecting different incremental configurations (e.g., single-class increments or grouped-class increments). The evaluated scenarios define the quantity of new classes introduced at each step $\mathcal{T}_i$:

- scenarios for the CICIDS-2017 dataset
  - 1+1+1+1+1+1+1
  - 3+4
  - 2+2+3
- scenarios for the UNSW-NB15 dataset
  - 1+1+1+1+1+1+1+1+1
  - 4+4
  - 2+2+4]

[1]A herding strategy is a greedy selection algorithm that selects m samples such that their average feature representation is as close as possible to the true mean of the entire class.

Training was performed sequentially over tasks. After completing each task, evaluation was conducted on all classes observed so far to measure forward transfer and forgetting. Model weights were saved after each task to ensure reproducibility.

Experiments were conducted using Google Colab with a Tesla T4 GPU and standard RAM configuration. Hyper-parameters such as batch size and memory buffer size were kept consistent across methods to ensure fair comparison.

4 Results

Performance of CIL could be evaluated from three point of views: overall performance of the tasks learned so far, memory stability of old tasks and learning plasticity of new tasks (Wang et al., 2024).

Due to the nature of intrusion detection systems, correctly identifying attack classes is more critical than classifying benign traffic. Therefore, we evaluate the model under two complementary perspectives: overall task accuracy and attack-focused accuracy.

To this end, we compute all CIL metrics using two evaluation matrices: the Accuracy Matrix (equation 1), which contains the classification accuracy over all classes for each task, and the Attack Accuracy Matrix (equation 2), which is computed by restricting the evaluation to attack samples only. This dual evaluation allows us to measure both general performance and security-critical behavior, particularly the model's ability to retain knowledge about malicious patterns over time.

$$A = [a_{k,j}] \tag{1}$$

$$A^{att} = [a^{att}_{k,j}] \tag{2}$$

- **Average Accuracy (AA):** Computed on the Accuracy Matrix A, it measures the mean classification accuracy over all learned tasks, considering both benign and attack classes.

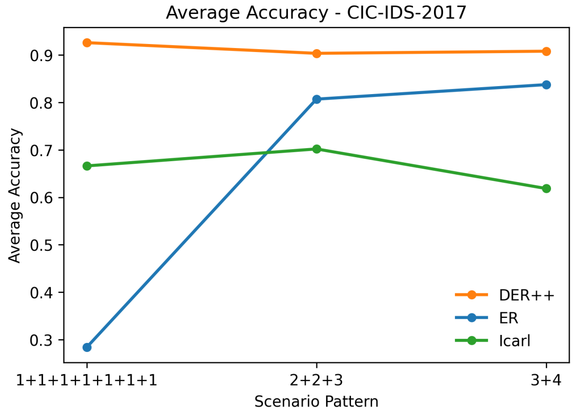
- **Average Attack Accuracy (AAA):** Computed on the Attack Accuracy Matrix A^{att}, it evaluates the mean accuracy over tasks when the evaluation is restricted to attack classes only, providing a security-oriented performance measure.

- **Average Forgetting (AF):** Computed on the Accuracy Matrix A, it is defined as the average degradation in performance on previously learned tasks after subsequent training steps.

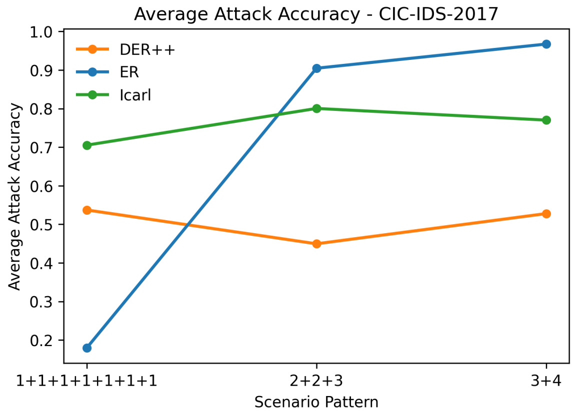
- **Average Attack Forgetting (AAF):** Computed on the Attack Accuracy Matrix A^{att}, it quantifies forgetting restricted to attack classes, highlighting the loss of previously learned malicious patterns.

Higher values of AA and AAA, indicate better knowledge retention and generalization, whereas lower values of AF and AAF reflect stronger resistance to catastrophic forgetting. Best results are highlighted according to these criteria. The experimental results are summarized in Table 1 and Figure 2 for the CICIDS-2017 dataset and in Table 2 and Figure 3 for the UNSW-NB15 dataset, where we compare the performance of iCaRL, DER++, and ER across different CIL scenarios. All methods were evaluated using the same model architecture, memory buffer size (5000), batch size (256), learning rate (0.001) and number of epochs (5) to ensure a fair comparison.

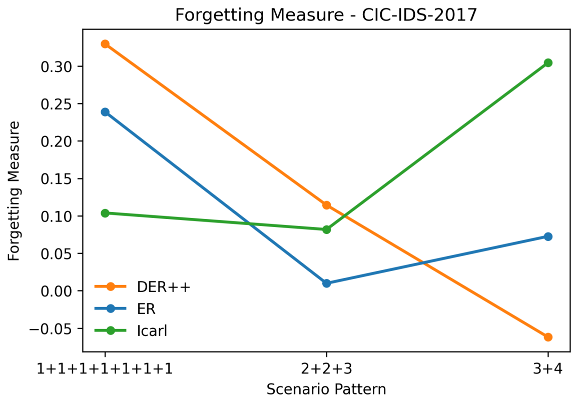
Across both CICIDS-2017 and UNSW-NB15, DER++ achieves the highest average accuracy in most scenarios, especially under highly incremental settings (e.g.,1+1+1+1+1+1+1+1), indicating better resistance to catasrophic forgetting. Since the memory buffer is fixed, these improvements are due to the learning strategy rather than additional replay resources. ER performs competitively when tasks are grouped into larger batches (e.g., 2+2+3 or 4+4), but its performance drops in highly fragmented scenarios, showing higher sensitivity to task interference. iCaRL maintains low attack forgetting in some fine-grained scenarios on CICIDS-2017, which suggests effective retention of previously learned attack classes through exemplar-based representations; however, it shows lower overall accuracy and a clear performance decrease on the more complex UNSW-NB15 dataset. In contrast, DER++ maintains more stable accuracy and lower or even negative forgetting in several settings, indicating better knowledge retention over time. Overall, the results show that, under a fixed memory budget, the choice of continual learning strategy has a larger impact than memory size, and DER++ provides the most consistent performance across datasets and scenario granularities.



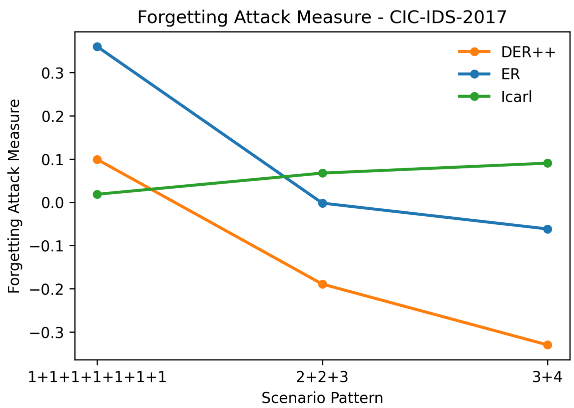
(a) Average Accuracy



(b) Average Attack Accuracy

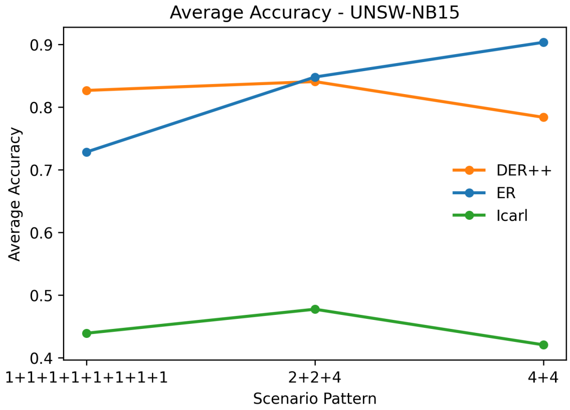


(c) Average Forgetting

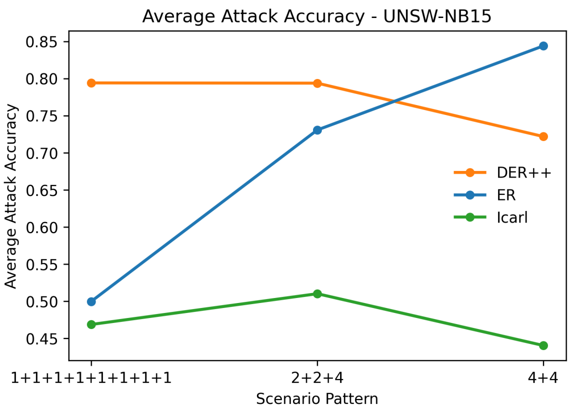


(d) Average Attack Forgetting

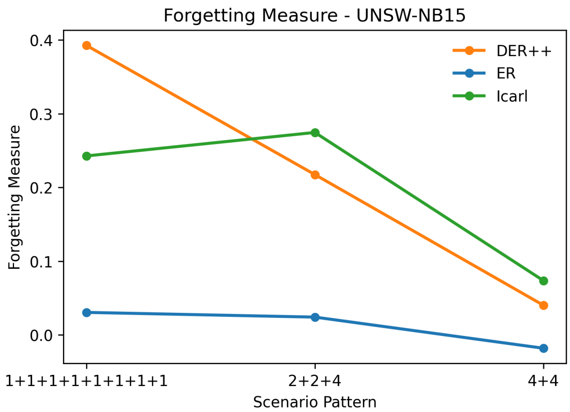
Figure 2: Results for dataset CICIDS-2017



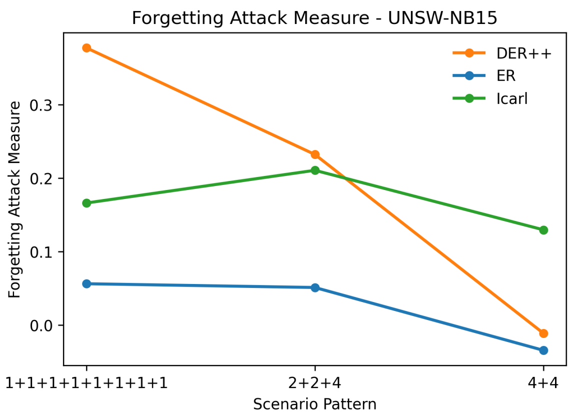
(a) Average Accuracy



(b) Average Attack Accuracy



(c) Average Forgetting



(d) Average Attack Forgetting

Figure 3: Results for dataset UNSW-NB15

Scenario	**Metric**	**iCaRL**	**DER++**	**ER**
1+1+1+1+1+1+1	AA	0.6662	**0.9261**	0.2836
	AAA	**0.7052**	0.5372	0.1794
	AF	**0.1039**	0.3297	0.2389
	AAF	**0.0186**	0.0990	0.3597
2+2+3	AA	0.7020	**0.9036**	0.8070
	AAA	0.8004	0.4495	**0.9046**
	AF	0.0818	0.1144	**0.0099**
	AAF	0.0675	**-0.1892**	-0.0020
3+4	AA	0.6185	**0.9081**	0.8376
	AAA	0.7702	0.5278	**0.9673**
	AF	0.3045	**-0.0618**	0.0727
	AAF	0.0904	**-0.3296**	-0.0616

Table 1: Results CIL Dataset CICIDS-2017

Scenario	**Metric**	**iCaRL**	**DER++**	**ER**
1+1+1+1+1+1+1+1	AA	0.4391	**0.8266**	0.7282
	AAA	0.4685	**0.7940**	0.4996
	AF	0.2428	0.3928	**0.0306**
	AAF	0.1661	0.3771	**0.0563**
2+2+4	AA	0.4774	0.8407	**0.8480**
	AAA	0.5100	**0.7937**	0.7307
	AF	0.2747	0.2175	**0.0243**
	AAF	0.2106	0.2321	**0.0512**
4+4	AA	0.4206	0.7837	**0.9034**
	AAA	0.4402	0.7218	**0.8440**
	AF	0.0737	0.0403	**-0.0180**
	AAF	0.1297	-0.0110	**-0.0342**

Table 2: Results CIL Dataset UNSW-NB15 (2015)

5 Conclusions

In this work, we proposed an adaptive Intrusion Detection System (IDS) using Class-Incremental Learning (CIL) to address the challenge of evolving cyber threats. Our experiments on the UNSW-NB15 and CICIDS-2017 datasets showed that Dark Experience Replay++ (DER++) outperforms both Experience Replay (ER) and Incremental Classifier and Representation Learning (iCaRL) in complex scenarios like the 1+1+1+1+1+1+1 sequence, where more forgetting occurs. DER++ was the most consistent, balancing stability and plasticity across tasks.

We also observed that the order in which attack categories are introduced significantly influences the results, underscoring the importance of task sequencing in CIL settings. However, in a real-world scenario is not possible to define a class ordering that minimizes forgetting and improves performance.

Overall, DER++ provides a reliable strategy for adaptive IDS, offering strong retention of previously learned attack patterns while accommodating the introduction of new classes, making it suitable for evolving cybersecurity challenges.

Future work could explore a hybrid continual learning framework that leverages the logit-preserving loss of DER++ together with the exemplar selection strategy of iCaRL. Specifically, replacing random reservoir sampling with herding-based memory construction may improve class representativeness within the fixed buffer, especially for minority attack classes. This approach could reduce Average Attack Forgetting by maintaining more informative and class-balanced exemplars, while still benefiting from the dark knowledge regularization that enables DER++ to achieve strong overall stability. This hybrid formulation is particularly suitable for intrusion detection systems, where class imbalance and the long-term retention of rare attack patterns are critical challenges.

6 Links to external resources

1. Repository
2. CICIDS-2017 dataset
3. UNSW-NB15 dataset

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